Guneesh Vats

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Applied AI researcher with experience in LLMs, multimodal learning, NLP, and agentic systems. Strong background in model development, benchmark design, experimental evaluation, and translating research into practical ML systems.

Education

International Institute of Information Technology (IIIT), Hyderabad *Aug 2020 – Present*
B.Tech and M.S. (Research), Electronics and Communication Engineering (ECE) MS GPA: **8.70/10** | B.Tech GPA: **8.00/10**
MS Thesis: Change-Point Detection via Structured Projections: From Channel Codes to Embeddings
Honors: Dean's List (Top 1% – Monsoon '21, Spring '23) | 7th globally, IEEE ICASSP Grand Challenge 2025 | Flipkart Grid 2023 Finalist | 3rd Globally, PAN @ CLEF Style Change Detection Challenge

Experience

Applied Research Scientist (ARS) – **ServiceNow** *Jan 2026 – Jun 2026 | Hyderabad*

- **REDACT – Multilingual PII Benchmark:** Built an EMNLP 2026 benchmark with **13,427 records and 324K annotations** across 51 entity types, 25 languages, and 9 scripts for controlled failure analysis.
- Engineered a **7-stage LLM data pipeline** for controlled generation, deterministic span alignment, verification, repair, semantic dedup, and release auditing; incorporated disclosure form and sensitivity metadata to support fine-grained evaluation.
- Evaluated **Presidio, GLiNER, OpenAI Privacy Filter & frontier LLMs** using exact, partial, and fuzzy span matching, exposing failures across language, entity type, disclosure form, and sensitivity tier.
- **ITSM Agent Benchmark:** Built a **13-stage synthetic-data pipeline** converting patterns from enterprise ITSM incidents into **1,000 synthetic incidents with 43 fields**, with normalization, clustering, stratified sampling, validation, repair, and re-validation.
- Architected a **3-tier ITSM evaluation framework** culminating in a simulated ServiceNow environment with **4 workflows, 7 ReAct agents, and 20 workflow-facing tools** for triage, retrieval, routing, resolution planning, action execution, and post-incident review.

Machine Learning Engineer Intern – **Coschool AI** *Jun 2025 – Dec 2025 | Hyderabad*

- **Question-level Multimodal Evaluation:** Built a handwritten-answer evaluation system with rubric-guided VLM scoring, achieving **95% accuracy**; direct-image evaluation reduced inference latency and cost by 35%.
- Cut per-request LLM input-token cost by **70%** ($0.0125 → $0.0038) via **prompt prefix caching** (53% to 87% cache hits using cyclic keys); reduced RAG prompt tokens by ~40% using cosine retrieval, **cross-encoder re-ranking**, and score filtering while maintaining grading quality.
- **Worksheet Segmentation Service:** Fine-tuned YOLOv11 (nano & small) for label/separator detection (**mAP50 0.975**) and deployed a Kubernetes-hosted FastAPI service supporting VLM question matching, cross-page answer stitching, and separator recovery.
- **Worksheet-level Batched Grading:** Evaluated every question in an answer script against its rubric in a single VLM call; benchmarked 8 GPT and 5 Claude models, **reducing input tokens by ~75% and evaluation cost by ~47%** without mis-mapped answers.
- Devised a Gaussian Mixture Model–based classifier to detect on- and off-topic queries in chatbot conversations with 95% accuracy.

Founding Machine Learning Engineer – **Alonzo AI** *Sep 2024 – Dec 2025 | Hyderabad*

- **Sports Intelligence CMS:** Architected a multimodal platform using fine-tuned YOLO and AWS Textract to digitize legacy record books, reconcile statistics against production schemas, and support source-linked review, correction, approval, and publishing.
- Developed a rare-event discovery and publishing layer over the structured sports database, combining rarity attributes, historical context, and record-level metadata to surface noteworthy performances and support direct publishing to social channels.
- **Conversational Sports Analytics:** Built a schema-aware engine that extracts entities, metrics, filters, and time constraints, maps them to supported query types, and populates validated SQL/MongoDB templates for execution in the CMS.
- Built a **schema-aware retrieval layer** with SBERT embeddings indexed in Milvus, combining metadata filtering, cross-encoder reranking, and score-based context pruning to retrieve relevant schemas and fields; **reduced query latency by 40% and LLM API cost by 30%**.

ML Engineer Intern (Speech) – **Adalat AI** *Apr 2024 – Jul 2024 | Remote*

- Reduced court **ASR word-error rate to 21.4% (18pp improvement)** on a 30K-minute multi-speaker dataset by fine-tuning Whisper via **LoRA/PEFT**, with PyAnnote diarization and log-mel alignment.
- Built a **real-time speech-recognition inference and model-serving backend** using FastAPI and NVIDIA Riva, achieving **~100 ms latency** through GPU caching, batching, token prefetching, and speaker-aware preprocessing.

Recent Publications

Complete list on Google Scholar.

- **REDACT: A Systematically Controlled Multilingual Benchmark for Personal Information Detection** – *EMNLP 2026, Accepted*
  Introduces a controlled multilingual benchmark to reveal where personal-information detectors miss or mislabel spans. Its 324K annotations cover 25 languages, 9 scripts, and 51 entity types, enabling failure analysis by language, entity type, disclosure form, and sensitivity.
- **Neural-Network Based Change-Point Detection for Binary Linear Block Codes** – *IEEE WCNC 2026, Published*
  Detects when the underlying binary linear block code changes in a received signal, enabling blind change-point detection. Combines structured projections with neural refinement to identify code transitions.
- **Detecting Convolutional Codes via a Markovian Statistic** – *IEEE WCNC 2026, Published*
  Enables blind identification of convolutional codes from received sequences, without relying on known code labels. Uses a Markovian statistic to capture temporal dependencies that distinguish code structures.
- **Decoding Emotion: Speech Perception Patterns in Individuals with Self-reported Depression** – *IEEE O-COCOSDA, 2024, Published*
  Studies whether self-reported depression is associated with differences in how listeners perceive emotion in speech, helping characterize depression-linked patterns in affective speech perception.
- **Benchmarking Agentic AI for Enterprise IT Service Management** – *ServiceNow Research, EMNLP 2026, Accepted*
  Evaluates whether AI agents can reliably manage enterprise IT incidents, from triage and classification through resolution planning and post-incident review. Uses a simulated ServiceNow environment with 4 workflows, 7 ReAct agents, 20 tools, 1,000 incidents, and 2,280 knowledge articles, assessed with 75+ metrics across 7 dimensions.
- **Contrastive Representation Learning for Authorship Change Detection** – *PAN @ CLEF 2026, Accepted*
  Identifies sentence-level authorship shifts in multi-author documents to locate where writing changes between contributors. Combines CoSENT-tuned DeBERTa-v3, RoBERTa-large, and ELECTRA-large with pairwise embedding features and MLP classifiers; Macro-F1 is 0.995, 0.920, and 0.810 on Easy, Medium, and Hard datasets.

Technical Skills

- **Languages & ML:** Python, C/C++, PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
- **LLMs & Retrieval:** LLM Fine-Tuning (LoRA/PEFT), LLM Evaluation, Prompt version control, RAG, Milvus, Pinecone, ChromaDB
- **Multimodal & Speech:** VLMs, YOLO, OCR/Document AI, Whisper, PyAnnote, NVIDIA Riva & NeMo
- **ML Systems & Serving:** FastAPI, Docker, Kubernetes, CUDA, ONNX, AWS (EC2, S3, SageMaker), GitHub Actions, Kubernetes, Flask
- **Data, Observability & Testing:** SQL, PostgreSQL, MongoDB, Redis, Neo4j, Prometheus, Grafana, pytest, JMeter, Locust, A/B Testing